Linear Statistical Models — Homework 1

B.Stat. 3rd year, Semester 1, 2026–27

Solutions

References to lecture notes. **[GM]** *Gauss–Markov Framework for Linear Models*; **[SLR]** *Overview of Simple Linear Regression*; **[PMI]** *Projection and Matrix Inversion.* Equation/result numbers below refer to these notes. Any result not stated there is proved.

Problem 1

Let $X \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let A be a symmetric $n \times n$ matrix.

A standardisation used in (b),(c). Since $\boldsymbol{\Sigma}$ is a covariance matrix it is symmetric positive semidefinite, so $\boldsymbol{\Sigma} = LL^\top$ for some $n \times n$ matrix L. If $U \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n)$, the affine image $\boldsymbol{\mu} + LU$ is Gaussian with mean $\boldsymbol{\mu}$ and covariance $L\,\mathrm{Var}(U)\,L^\top = LL^\top = \boldsymbol{\Sigma}$; hence $\boldsymbol{\mu} + LU \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. We may therefore take

$$X = \boldsymbol{\mu} + LU, \qquad U \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n).$$

For i.i.d. standard normal coordinates $U_1, \dots, U_n$ we use the elementary moments $\mathbb{E}U_i = 0$, $\mathbb{E}U_i^2 = 1$, $\mathbb{E}U_i^3 = 0$, $\mathbb{E}U_i^4 = 3$ and independence, which give

$$\mathbb{E}[U_iU_j] = \delta_{ij}, \tag{1}$$
$$\mathbb{E}[U_iU_jU_k] = 0, \tag{2}$$
$$\mathbb{E}[U_iU_jU_kU_\ell] = \delta_{ij}\delta_{k\ell} + \delta_{ik}\delta_{j\ell} + \delta_{i\ell}\delta_{jk}. \tag{3}$$

Indeed (2) holds because at least one index is unpaired (its factor contributes $\mathbb{E}U_i = 0$), and (3) is checked by cases: all four indices equal gives $\mathbb{E}U_i^4 = 3 = 1 + 1 + 1$; two distinct pairs gives 1; otherwise an unpaired index gives 0 — exactly what the right-hand side returns.

(a) $\mathbb{E}(X^\top AX) = \mathrm{tr}(A\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top A\boldsymbol{\mu}$

This needs only $\mathbb{E}X = \boldsymbol{\mu}$ and $\mathrm{Var}(X) = \boldsymbol{\Sigma}$. From $\boldsymbol{\Sigma} = \mathbb{E}[(X - \boldsymbol{\mu})(X - \boldsymbol{\mu})^\top] = \mathbb{E}[XX^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top$ we get $\mathbb{E}[XX^\top] = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top$. Since $X^\top AX$ is a scalar, $X^\top AX = \mathrm{tr}(X^\top AX) = \mathrm{tr}(AXX^\top)$, and by linearity of tr and $\mathbb{E}$,

$$\mathbb{E}(X^\top AX) = \mathrm{tr}\big(A\,\mathbb{E}[XX^\top]\big) = \mathrm{tr}\big(A(\boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top)\big) = \mathrm{tr}(A\boldsymbol{\Sigma}) + \mathrm{tr}(\boldsymbol{\mu}^\top A\boldsymbol{\mu}) = \mathrm{tr}(A\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top A\boldsymbol{\mu}. \qquad \blacksquare$$

(b) $\mathrm{Cov}(X, X^\top AX) = 2\boldsymbol{\Sigma}A\boldsymbol{\mu}$

Write $X = \boldsymbol{\mu} + LU$. Using symmetry of A ($\boldsymbol{\mu}^\top ALU = U^\top L^\top A\boldsymbol{\mu}$, both scalars),

$$X^\top AX = \boldsymbol{\mu}^\top A\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top ALU + U^\top \underbrace{L^\top AL}_{=:B}\, U.$$

The constant $\boldsymbol{\mu}^\top A\boldsymbol{\mu}$ does not affect covariance, and $X - \mathbb{E}X = LU$, so

$$\mathrm{Cov}(X, X^\top AX) = \mathrm{Cov}\big(LU,\ 2(L^\top A\boldsymbol{\mu})^\top U\big) + \mathrm{Cov}\big(LU,\ U^\top BU\big).$$

With $v := L^\top A\boldsymbol{\mu}$, the first term is $2L\,\mathbb{E}[UU^\top]\,v = 2Lv = 2LL^\top A\boldsymbol{\mu} = 2\boldsymbol{\Sigma}A\boldsymbol{\mu}$ by (1). The i-th component of the second term is $\sum_{j,k} B_{jk}\,\mathbb{E}[U_iU_jU_k] = 0$ by (2). Hence $\mathrm{Cov}(X, X^\top AX) = 2\boldsymbol{\Sigma}A\boldsymbol{\mu}$. $\blacksquare$

(c) $\operatorname{Var}(X^\top AX) = 2\operatorname{tr}\big((A\boldsymbol{\Sigma})^2\big) + 4\boldsymbol{\mu}^\top A\boldsymbol{\Sigma}A\boldsymbol{\mu}$

From (b), $X^\top AX = \text{const} + 2v^\top U + U^\top BU$ with $v = L^\top A\boldsymbol{\mu}$, $B = L^\top AL$. Thus

$$\operatorname{Var}(X^\top AX) = 4\operatorname{Var}(v^\top U) + \operatorname{Var}(U^\top BU) + 4\operatorname{Cov}(v^\top U,\ U^\top BU).$$

Linear part. $\operatorname{Var}(v^\top U) = v^\top \operatorname{Var}(U)v = v^\top v = \boldsymbol{\mu}^\top ALL^\top A\boldsymbol{\mu} = \boldsymbol{\mu}^\top A\boldsymbol{\Sigma}A\boldsymbol{\mu}$, so $4\operatorname{Var}(v^\top U) = 4\boldsymbol{\mu}^\top A\boldsymbol{\Sigma}A\boldsymbol{\mu}$.

Cross term. $\operatorname{Cov}(v^\top U, U^\top BU) = \sum_i v_i \sum_{j,k} B_{jk}\,\mathbb{E}[U_iU_jU_k] = 0$ by (2).

Quadratic part. $\mathbb{E}[U^\top BU] = \sum_i B_{ii} = \operatorname{tr} B$, and by (3),

$$\mathbb{E}\big[(U^\top BU)^2\big] = \sum_{i,j,k,\ell} B_{ij}B_{k\ell}\,\mathbb{E}[U_iU_jU_kU_\ell] = (\operatorname{tr} B)^2 + \sum_{i,j} B_{ij}^2 + \sum_{i,j} B_{ij}B_{ji} = (\operatorname{tr} B)^2 + 2\operatorname{tr}(B^2),$$

where the last equality uses $B^\top = B$. Hence $\operatorname{Var}(U^\top BU) = 2\operatorname{tr}(B^2)$. Finally, since $LL^\top = \boldsymbol{\Sigma}$,

$$\operatorname{tr}(B^2) = \operatorname{tr}(L^\top AL\,L^\top AL) = \operatorname{tr}(A\,LL^\top A\,LL^\top) = \operatorname{tr}\big((A\boldsymbol{\Sigma})^2\big).$$

Collecting the three parts,

$$\operatorname{Var}(X^\top AX) = 4\boldsymbol{\mu}^\top A\boldsymbol{\Sigma}A\boldsymbol{\mu} + 2\operatorname{tr}\big((A\boldsymbol{\Sigma})^2\big). \qquad \blacksquare$$

Problem 2

$X \sim \mathcal{N}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite; A, B symmetric. *Claim:* $X^\top AX$ and BX are independent $\iff B\boldsymbol{\Sigma}A = \mathbf{0}$.

Solution. Because $\boldsymbol{\Sigma}$ is positive definite we may write $\boldsymbol{\Sigma} = LL^\top$ with L *invertible* (e.g. a Cholesky factor), and put $X = \boldsymbol{\mu} + LU$, $U \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n)$. Set

$$M := BL, \qquad P := L^\top AL \text{ (symmetric)}, \qquad a := L^\top A\boldsymbol{\mu}.$$

Then $BX = B\boldsymbol{\mu} + MU$ and $X^\top AX = c + 2a^\top U + U^\top PU$ with $c = \boldsymbol{\mu}^\top A\boldsymbol{\mu}$. Note

$$MP = BL\,L^\top AL = B\boldsymbol{\Sigma}A\,L, \qquad Ma = BL\,L^\top A\boldsymbol{\mu} = B\boldsymbol{\Sigma}A\boldsymbol{\mu},$$

so, as L is invertible, $MP = \mathbf{0} \iff B\boldsymbol{\Sigma}A = \mathbf{0}$, and this implies $Ma = B\boldsymbol{\Sigma}A\boldsymbol{\mu} = \mathbf{0}$.

Joint MGF. For $z \sim \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n)$ and any vector b and symmetric Q with $\mathbf{I} - 2Q$ positive definite,

$$\mathbb{E}\exp\big(b^\top z + z^\top Qz\big) = \int \frac{e^{-\frac{1}{2}z^\top(\mathbf{I}-2Q)z + b^\top z}}{(2\pi)^{n/2}}\,dz = \det(\mathbf{I} - 2Q)^{-1/2}\exp\Big(\tfrac{1}{2}b^\top(\mathbf{I} - 2Q)^{-1}b\Big),$$

by completing the square. Apply this with $s \in \mathbb{R}^n$, $t \in \mathbb{R}$ small (so $\mathbf{I} - 2tP \succ 0$): since $s^\top BX + t\,X^\top AX = (s^\top B\boldsymbol{\mu} + tc) + (M^\top s + 2ta)^\top U + t\,U^\top PU$, writing $N := (\mathbf{I} - 2tP)^{-1}$ and $b := M^\top s + 2ta$,

$$\varphi(s,t) := \mathbb{E}\,e^{s^\top BX + tX^\top AX} = e^{s^\top B\boldsymbol{\mu} + tc}\,\det(\mathbf{I} - 2tP)^{-1/2}\exp\big(\tfrac{1}{2}b^\top Nb\big).$$

The marginals are $\varphi(s,0) = e^{s^\top B\boldsymbol{\mu}}\exp(\frac{1}{2}s^\top MM^\top s)$ and $\varphi(0,t) = e^{tc}\det(\mathbf{I} - 2tP)^{-1/2}\exp(2t^2a^\top Na)$. Since the MGF is finite on a neighbourhood of the origin, BX and $X^\top AX$ are independent *iff* $\varphi(s,t) = \varphi(s,0)\varphi(0,t)$ there, i.e. iff $\frac{1}{2}b^\top Nb = \frac{1}{2}s^\top MM^\top s + 2t^2 a^\top Na$. Expanding $b^\top Nb = s^\top MNM^\top s + 4t\,s^\top MNa + 4t^2 a^\top Na$ and cancelling $2t^2a^\top Na$, the condition becomes

$$\tfrac{1}{2}\,s^\top MNM^\top s + 2t\,s^\top MNa = \tfrac{1}{2}\,s^\top MM^\top s \qquad \text{for all } s \in \mathbb{R}^n,\ t \text{ near } 0. \tag{$\star$}$$

For fixed t, ($\star$) holds for all s iff its quadratic and linear parts in s match separately (replace $s \mapsto -s$), i.e.

$$MNM^\top = MM^\top \quad \text{and} \quad MNa = \mathbf{0}, \qquad N = (\mathbf{I} - 2tP)^{-1}.$$

Using the convergent expansion $N = \sum_{k\ge 0}(2t)^k P^k$ for small t:

$$MNM^\top - MM^\top = \sum_{k\ge 1}(2t)^k MP^kM^\top = \mathbf{0}\ \forall t \iff MP^kM^\top = \mathbf{0}\ \forall k \ge 1.$$

Taking $k = 2$ gives $MP^2M^\top = (MP)(MP)^\top = \mathbf{0}$, and $CC^\top = \mathbf{0} \Rightarrow C = \mathbf{0}$ (its diagonal entries are squared row norms); hence $MP = \mathbf{0}$. Conversely $MP = \mathbf{0}$ gives $MP^kM^\top = \mathbf{0}$ for all $k \ge 1$ and $MN = M$, so also $MNa = Ma$. Thus ($\star$) holds for all s, t iff

$$MP = \mathbf{0} \ \text{ and } \ Ma = \mathbf{0}.$$

Finally $MP = \mathbf{0} \iff B\boldsymbol{\Sigma}A = \mathbf{0}$, and $B\boldsymbol{\Sigma}A = \mathbf{0}$ already forces $Ma = B\boldsymbol{\Sigma}A\boldsymbol{\mu} = \mathbf{0}$. Therefore

$$X^\top AX \text{ and } BX \text{ are independent} \iff B\boldsymbol{\Sigma}A = \mathbf{0}. \qquad \blacksquare$$

Remark. The "if" direction has a shorter argument: $B\boldsymbol{\Sigma}A = \mathbf{0} \Rightarrow A\boldsymbol{\Sigma}B = (B\boldsymbol{\Sigma}A)^\top = \mathbf{0}$, so $\mathrm{Cov}(AX, BX) = A\boldsymbol{\Sigma}B^\top = \mathbf{0}$; jointly normal and uncorrelated $\Rightarrow AX \perp BX$; and $X^\top AX = (AX)^\top A^+(AX)$ (using the Moore–Penrose inverse A^+, which exists by [PMI, Prop. 15], and $AA^+A = A$) is a function of AX, hence independent of BX. The MGF computation is needed for "only if".

Problem 3

Model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\mathbf{x}_i = (1, x_i)^\top$, $\mathbf{X}$ the $n \times 2$ design with rows $\mathbf{x}_i^\top$, and $\boldsymbol{\beta} = (\beta_0, \beta_1)^\top$. Assume $\mathbf{X}$ has full column rank, so $\mathbf{X}^\top\mathbf{X}$ is nonsingular and $\widehat{\boldsymbol{\beta}}^{(n)} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ [GM, full-rank LSE]. Write $G := (\mathbf{X}^\top\mathbf{X})^{-1}$ and $\mathbf{u} := \mathbf{x}_{n+1}$.

A rank-one inverse update (Sherman–Morrison). For nonsingular M and a vector $\mathbf{u}$ with $1 + \mathbf{u}^\top M^{-1}\mathbf{u} \neq 0$,

$$(M + \mathbf{u}\mathbf{u}^\top)^{-1} = M^{-1} - \frac{M^{-1}\mathbf{u}\mathbf{u}^\top M^{-1}}{1 + \mathbf{u}^\top M^{-1}\mathbf{u}}. \tag{SM}$$

Proof: multiply the right-hand side by $(M + \mathbf{u}\mathbf{u}^\top)$ and let $\gamma := 1 + \mathbf{u}^\top M^{-1}\mathbf{u}$:

$$\begin{aligned}
&(M + \mathbf{u}\mathbf{u}^\top)\Big(M^{-1} - \tfrac{1}{\gamma}M^{-1}\mathbf{u}\mathbf{u}^\top M^{-1}\Big)\\
&\quad = \mathbf{I} + \mathbf{u}\mathbf{u}^\top M^{-1} - \tfrac{1}{\gamma}\mathbf{u}\mathbf{u}^\top M^{-1} - \tfrac{1}{\gamma}\mathbf{u}(\mathbf{u}^\top M^{-1}\mathbf{u})\mathbf{u}^\top M^{-1}\\
&\quad = \mathbf{I} + \mathbf{u}\mathbf{u}^\top M^{-1}\Big(1 - \tfrac{\gamma}{\gamma}\Big) = \mathbf{I}. \quad \checkmark
\end{aligned}$$

Here $M = \mathbf{X}^\top\mathbf{X}$ is positive definite, so $\mathbf{u}^\top G\mathbf{u} \ge 0$ and $\gamma = 1 + \mathbf{u}^\top G\mathbf{u} \ge 1 > 0$; thus $c_{n+1} = (1 + \mathbf{x}_{n+1}^\top G\,\mathbf{x}_{n+1})^{-1} \in (0, 1]$ is well defined.

(a)

Adding (x_{n+1}, Y_{n+1}) stacks one row $\mathbf{x}_{n+1}^\top$ on $\mathbf{X}$ and one entry Y_{n+1} on $\mathbf{Y}$, so

$$\mathbf{X}_{n+1}^\top \mathbf{X}_{n+1} = \mathbf{X}^\top \mathbf{X} + \mathbf{u}\mathbf{u}^\top, \qquad \mathbf{X}_{n+1}^\top \mathbf{Y}_{(n+1)} = \mathbf{X}^\top \mathbf{Y} + \mathbf{u}\, Y_{n+1}.$$

By (SM) with $M = \mathbf{X}^\top\mathbf{X}$, and writing $c := c_{n+1}$, $(\mathbf{X}_{n+1}^\top \mathbf{X}_{n+1})^{-1} = G - c\, G\mathbf{u}\mathbf{u}^\top G$. Hence

$$\widehat{\boldsymbol{\beta}}^{(n+1)} = (G - c\, G\mathbf{u}\mathbf{u}^\top G)(\mathbf{X}^\top\mathbf{Y} + \mathbf{u}Y_{n+1}) = \widehat{\boldsymbol{\beta}}^{(n)} + G\mathbf{u}\, Y_{n+1} - c\, G\mathbf{u}\, \mathbf{u}^\top \widehat{\boldsymbol{\beta}}^{(n)} - c\, G\mathbf{u}(\mathbf{u}^\top G\mathbf{u})Y_{n+1},$$

using $G\mathbf{X}^\top\mathbf{Y} = \widehat{\boldsymbol{\beta}}^{(n)}$ and $\mathbf{u}^\top G\mathbf{X}^\top\mathbf{Y} = \mathbf{u}^\top\widehat{\boldsymbol{\beta}}^{(n)}$. Now $\mathbf{u}^\top\widehat{\boldsymbol{\beta}}^{(n)} = \mathbf{x}_{n+1}^\top\widehat{\boldsymbol{\beta}}^{(n)} = \widehat{Y}_{n+1}^{(n)}$ (the value predicted for Y_{n+1} from the first n observations — this is the BLUE of $\mathbb{E}Y_{n+1} = \mathbf{x}_{n+1}^\top\boldsymbol{\beta}$, [GM, Thm 1]), and $c(\mathbf{u}^\top G\mathbf{u}) = c(\frac{1}{c} - 1) = 1 - c$. Factoring out $G\mathbf{u}$,

$$\widehat{\boldsymbol{\beta}}^{(n+1)} - \widehat{\boldsymbol{\beta}}^{(n)} = G\mathbf{u}\Big[\, Y_{n+1}\big(1 - c\, \mathbf{u}^\top G\mathbf{u}\big) - c\, \widehat{Y}_{n+1}^{(n)}\Big] = G\mathbf{u}\Big[c\, Y_{n+1} - c\, \widehat{Y}_{n+1}^{(n)}\Big],$$

because $1 - c\, \mathbf{u}^\top G\mathbf{u} = 1 - (1 - c) = c$. Therefore

$$\boxed{\widehat{\boldsymbol{\beta}}^{(n+1)} - \widehat{\boldsymbol{\beta}}^{(n)} = c_{n+1}\big(Y_{n+1} - \widehat{Y}_{n+1}^{(n)}\big)(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_{n+1}.} \qquad \blacksquare$$

(b)

Left-multiply the boxed identity by $\mathbf{x}_{n+1}^\top = \mathbf{u}^\top$. Since $\widehat{Y}_{n+1}^{(n+1)} = \mathbf{x}_{n+1}^\top\widehat{\boldsymbol{\beta}}^{(n+1)}$ and $\widehat{Y}_{n+1}^{(n)} = \mathbf{u}^\top\widehat{\boldsymbol{\beta}}^{(n)}$,

$$\widehat{Y}_{n+1}^{(n+1)} - \widehat{Y}_{n+1}^{(n)} = c_{n+1}\big(Y_{n+1} - \widehat{Y}_{n+1}^{(n)}\big)\, \mathbf{u}^\top G\mathbf{u}.$$

Subtract both sides from $Y_{n+1} - \widehat{Y}_{n+1}^{(n)}$ and use $1 - c_{n+1}\, \mathbf{u}^\top G\mathbf{u} = c_{n+1}$:

$$Y_{n+1} - \widehat{Y}_{n+1}^{(n+1)} = \big(Y_{n+1} - \widehat{Y}_{n+1}^{(n)}\big)\big(1 - c_{n+1}\mathbf{u}^\top G\mathbf{u}\big) = c_{n+1}\big(Y_{n+1} - \widehat{Y}_{n+1}^{(n)}\big).$$

Dividing by $c_{n+1} \in (0, 1]$,

$$Y_{n+1} - \widehat{Y}_{n+1}^{(n)} = \frac{Y_{n+1} - \widehat{Y}_{n+1}^{(n+1)}}{c_{n+1}}. \qquad \blacksquare$$

Problem 4

Normal simple linear regression, $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$; test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$. Put $S_{xx} := \sum_i (x_i - \bar{x})^2$.

Remark. With two parameters (β_0, β_1) the residual degrees of freedom are $n - 2$. The correct statement — matching [SLR, eq. (12)] and part (b) below — is $T \sim t_{n-2}$; the "$t_{(n-1)}$" in the problem is a typo for $t_{(n-2)}$. We prove $T \sim t_{n-2}$ under H_0.

(a) $T = \widehat{\beta}_1/s(\widehat{\beta}_1) \sim t_{n-2}$ under H_0

Write $\widehat{\beta}_1 = \sum_i c_i Y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$ [SLR, eq. (4)]. Then $\sum_i c_i = 0$ and $\sum_i c_i x_i = \sum_i (x_i - \bar{x}) x_i / S_{xx} = 1$, so $\mathbb{E}\widehat{\beta}_1 = \sum_i c_i(\beta_0 + \beta_1 x_i) = \beta_1$ and $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2 \sum_i c_i^2 = \sigma^2/S_{xx}$ [SLR, eq. (7)]. Being a linear combination of independent normals, $\widehat{\beta}_1$ is normal:

$$\widehat{\beta}_1 \sim \mathcal{N}\Big(\beta_1,\ \tfrac{\sigma^2}{S_{xx}}\Big), \qquad \text{so under } H_0: \quad Z := \frac{\widehat{\beta}_1 \sqrt{S_{xx}}}{\sigma} \sim \mathcal{N}(0,1).$$

By [SLR, eq. (12)], SSE $\sim \sigma^2 \chi^2_{n-2}$ and SSE is independent of $\widehat{\beta}_1$; write $W := \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-2}$, independent of Z. Since $s^2(\widehat{\beta}_1) = \mathrm{MSE}/S_{xx}$ with $\mathrm{MSE} = \mathrm{SSE}/(n-2)$ [SLR, eq. (8)],

$$T = \frac{\widehat{\beta}_1}{s(\widehat{\beta}_1)} = \frac{\widehat{\beta}_1\sqrt{S_{xx}}}{\sqrt{\mathrm{MSE}}} = \frac{\widehat{\beta}_1\sqrt{S_{xx}}/\sigma}{\sqrt{\mathrm{SSE}/\big((n-2)\sigma^2\big)}} = \frac{Z}{\sqrt{W/(n-2)}}.$$

A ratio $Z/\sqrt{W/k}$ with $Z \sim \mathcal{N}(0,1) \perp W \sim \chi^2_k$ is, by definition, t_k. Hence under H_0, $T \sim t_{n-2}$. ■

(b) $F = \mathrm{MSR}/\mathrm{MSE} = T^2$

The ANOVA decomposition SSTO = SSR + SSE has d.f.(SSR) = 1 with $\mathrm{SSR} = \widehat{\beta}_1^2 S_{xx}$ [SLR, ANOVA]. Thus $\mathrm{MSR} = \mathrm{SSR}/1 = \widehat{\beta}_1^2 S_{xx}$ and, since $s^2(\widehat{\beta}_1) = \mathrm{MSE}/S_{xx}$,

$$F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{\widehat{\beta}_1^2 S_{xx}}{\mathrm{MSE}} = \frac{\widehat{\beta}_1^2}{\mathrm{MSE}/S_{xx}} = \frac{\widehat{\beta}_1^2}{s^2(\widehat{\beta}_1)} = \Big(\frac{\widehat{\beta}_1}{s(\widehat{\beta}_1)}\Big)^2 = T^2. \qquad ■$$

(c) The two tests always agree

If $T \sim t_{n-2}$ then $T^2 = Z^2/(W/(n-2))$ with $Z^2 \sim \chi^2_1 \perp W \sim \chi^2_{n-2}$, so $T^2 \sim F_{1,n-2}$ (consistent with (b)). For the level-α tests:

$$t\text{-test rejects} \iff |T| > t_{n-2}(1-\tfrac{\alpha}{2}), \qquad F\text{-test rejects} \iff F = T^2 > F_{1,n-2}(1-\alpha).$$

Let $c := t_{n-2}(1-\frac{\alpha}{2})$. Then

$$\Pr\big(F_{1,n-2} > c^2\big) = \Pr(T^2 > c^2) = \Pr(|T| > c) = \Pr(T > c) + \Pr(T < -c) = \tfrac{\alpha}{2} + \tfrac{\alpha}{2} = \alpha,$$

so c^2 is exactly the upper-α point of $F_{1,n-2}$, i.e. $F_{1,n-2}(1-\alpha) = c^2$. Consequently

$$|T| > c \iff T^2 > c^2 \iff F > F_{1,n-2}(1-\alpha),$$

so for every data set and every α the two rejection regions coincide; the tests reach identical conclusions. ■

Problem 5

$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ with rows $\mathbf{x}_i^\top = (1, x_i)$; $h_{ii} = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$ is the i-th diagonal of $\mathbf{H}$. Let $\widehat{Y}_i = \mathbf{x}_i^\top\widehat{\boldsymbol{\beta}}$ (full-data fit) and $\widehat{Y}_i^{(-i)}$ the fit at x_i from the regression that omits observation i. *Claim:* $Y_i - \widehat{Y}_i^{(-i)} = (Y_i - \widehat{Y}_i)/(1 - h_{ii})$.

Solution. Apply Problem 3 with the roles reversed: take the "base" data set to be the $n-1$ observations $\{(x_j, Y_j)\}_{j\neq i}$ (design $\mathbf{X}_{(-i)}$, assumed full rank) and let observation i play the role of the "$(n+1)$-th" added point. In Problem 3's notation this identifies

$$\widehat{\boldsymbol{\beta}}^{(n)} \to \widehat{\boldsymbol{\beta}}^{(-i)}, \quad \widehat{\boldsymbol{\beta}}^{(n+1)} \to \widehat{\boldsymbol{\beta}}, \quad \widehat{Y}_{n+1}^{(n)} \to \widehat{Y}_i^{(-i)}, \quad \widehat{Y}_{n+1}^{(n+1)} \to \widehat{Y}_i, \quad c_{n+1} \to c := \big(1 + \mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i\big)^{-1},$$

where $M := \mathbf{X}_{(-i)}^{\mathsf{T}} \mathbf{X}_{(-i)}$. Problem 3(b) then reads

$$Y_i - \widehat{Y}_i^{(-i)} = \frac{Y_i - \widehat{Y}_i}{c}. \tag{5.1}$$

It remains to show $c = 1 - h_{ii}$. Deleting row i gives $\mathbf{X}^{\mathsf{T}}\mathbf{X} = M + \mathbf{x}_i \mathbf{x}_i^{\mathsf{T}}$, so with $g := \mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i$ and (SM) from Problem 3,

$$h_{ii} = \mathbf{x}_i^{\mathsf{T}} (\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1} \mathbf{x}_i = \mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i - \frac{(\mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i)^2}{1 + \mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i} = g - \frac{g^2}{1+g} = \frac{g}{1+g}.$$

Therefore $1 - h_{ii} = \dfrac{1}{1+g} = \big(1 + \mathbf{x}_i^{\mathsf{T}} M^{-1} \mathbf{x}_i\big)^{-1} = c$. Substituting into (5.1),

$$Y_i - \widehat{Y}_i^{(-i)} = \frac{Y_i - \widehat{Y}_i}{1 - h_{ii}}, \qquad i = 1, \dots, n. \qquad \blacksquare$$

Problem 6

$\mathcal{M}_1 : \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $\mathcal{M}_2 : \mathbf{Y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\eta}$, both with mean-zero errors of covariance $\sigma^2\mathbf{I}$, $\mathbf{Z}$ of full column rank. Being a *reparametrisation* means the two models describe the same mean vector, i.e. $\{\mathbf{X}\boldsymbol{\beta}\} = \{\mathbf{Z}\boldsymbol{\theta}\}$, so the model spaces coincide: $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{Z}) =: \mathcal{M}$.

Solution. *BLUE of* $\mathbf{X}\boldsymbol{\beta}$*.* Each component $(\mathbf{X}\boldsymbol{\beta})_i = \mathbf{x}_i^{\mathsf{T}}\boldsymbol{\beta}$ has coefficient vector $\mathbf{x}_i$ (the i-th row of $\mathbf{X}$) lying in $\mathcal{C}(\mathbf{X}^{\mathsf{T}}) = \mathcal{R}(\mathbf{X})$, hence is estimable [GM, Lemma 1]; by the Gauss–Markov theorem [GM, Thm 1] its BLUE is the corresponding component of the (unique) fitted vector

$$\widehat{\mathbf{X}\boldsymbol{\beta}} = \mathbf{P}_{\mathcal{M}}\, \mathbf{Y},$$

the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{M} = \mathcal{C}(\mathbf{X})$ [GM, eq. (2)]. The orthogonal projector onto a subspace is unique [PMI, Prop. 4], so it is the same whether computed from $\mathbf{X}$ or $\mathbf{Z}$: $\mathbf{P}_{\mathcal{M}} = \mathbf{X}(\mathbf{X}^{\mathsf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathsf{T}} = \mathbf{Z}(\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}$ (the latter valid since $\mathbf{Z}$ has full column rank, [PMI, Prop. 5]).

BLUE of $\boldsymbol{\theta}$*.* As $\mathbf{Z}$ has full column rank, every $\boldsymbol{\theta}_j = \mathbf{e}_j^{\mathsf{T}}\boldsymbol{\theta}$ is estimable and, by the Gauss–Markov theorem, the BLUE of $\boldsymbol{\theta}$ is the least-squares estimator under $\mathcal{M}_2$,

$$\widehat{\boldsymbol{\theta}} = (\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{Y}.$$

Moreover $\mathbf{Z}\widehat{\boldsymbol{\theta}} = \mathbf{Z}(\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{Y} = \mathbf{P}_{\mathcal{M}}\mathbf{Y}$.

The two BLUEs in terms of each other. Combining the displays,

$$\boxed{\widehat{\mathbf{X}\boldsymbol{\beta}} = \mathbf{P}_{\mathcal{M}}\mathbf{Y} = \mathbf{Z}\widehat{\boldsymbol{\theta}}} \qquad \text{and} \qquad \boxed{\widehat{\boldsymbol{\theta}} = (\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\, \widehat{\mathbf{X}\boldsymbol{\beta}}.}$$

The second identity is consistent since, using $\mathbf{P}_{\mathcal{M}}\mathbf{Y} = \mathbf{Z}\widehat{\boldsymbol{\theta}}$,

$$\begin{aligned}(\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{P}_{\mathcal{M}}\mathbf{Y} &= (\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{Z}(\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{Y} \\ &= (\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}\mathbf{Y} = \widehat{\boldsymbol{\theta}}.\end{aligned}$$

Thus the BLUE of $\mathbf{X}\boldsymbol{\beta}$ is $\mathbf{Z}$ times the BLUE of $\boldsymbol{\theta}$, and the BLUE of $\boldsymbol{\theta}$ is recovered from that of $\mathbf{X}\boldsymbol{\beta}$ by the left inverse $(\mathbf{Z}^{\mathsf{T}}\mathbf{Z})^{-1}\mathbf{Z}^{\mathsf{T}}$ of $\mathbf{Z}$. $\blacksquare$